

Project Catalogue

Eight themes, P01–P08

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Applied Machine Learning (CSAI2017P) — Project Catalogue, Autumn 2026

This file is the overview. **The rules are in `Project-Handbook.md`; the full brief for each theme is in its own folder** (P01-bike-demand/, P02-air-quality/, ...). Read the handbook, then the folder for the theme you pick.

How the component works

Sub-component	Marks	Granularity
Project quiz — a written paper, one set per theme	10	per student
Repository submission — state and commit history of the group's GitHub repo	5	per group, identical for every member

Read the weighting before you choose a group. The quiz is worth twice the repo mark, and the repo mark is **not** divided by contribution — every member of a group receives the same 5. A passenger therefore collects the group marks and then sits a paper about work they did not do; a contributor in a weak group can still score well. Every member sits the paper separately.

Groups: 4–8 students, self-formed. **At most two groups per theme**, first come, first served on the sign-up sheet. Themes P01–P08.

You submit one thing: the GitHub repository URL. No zip, no attachment. I read the commit history — created in Unit IV, at least 20 commits across at least 6 distinct weeks, no more than 40 % of them in the final week, every member committing under their own account, and both checkpoint tags pushed on time. A repository that appears in 1–3 commits in the last few days before submission scores at most 1.5 of 5, however good the code inside it is. Full conditions in `Project-Handbook.md` §3.

Timeline (anchored to units, not dates)

Milestone	Anchored to
Catalogue released; groups formed; theme selected; repo created	during / end of Unit IV
Checkpoint 1 — data loaded and shown, one baseline number (tag checkpoint-1)	end of Unit V
Checkpoint 2 — full pipeline, initial results (tag checkpoint-2)	end of Unit VI
Final submission — the repository URL	end of Unit VII
Project quiz — written, individual	final lecture / final lab (Lab experiment 15 slot)

Data on campus. The UPES network breaks Python downloads (`fetch_*`, `ucimlrepo`, `yfinance`, `requests`) — see `../lab-assignments/Anchor-Datasets.pdf`. Every theme below therefore uses a file you **download once in a browser** (or at home). Files under 50 MB are committed to `data/raw/`; the two larger ones (P05, P08) are gitignored with download instructions in the README — your theme brief says which applies. Do not disable certificate verification.

Common deliverables for every theme: a Git repository with `README.md`, `requirements.txt`, notebooks/scripts that run top-to-bottom from a clean checkout, a 4–6 page report (problem, data, method, results, limitations), and reproducibility notes (random seeds, environment, exact commands). The exact layout is in `Project-Handbook.md` §3.4.

The eight themes

Each theme is stated as a question, not a technique. The “done” line is the minimum for full group marks; the “further” items are where a strong group separates itself. **The summaries below are not the brief** — open the theme’s folder for the phase-by-phase instructions, the pitfalls and the “done” checklist.

Theme	Folder
P01	P01-bike-demand/
P02	P02-air-quality/
P03	P03-hotel-cancellations/
P04	P04-student-dropout/
P05	P05-review-sentiment/
P06	P06-fashion-mnist/
P07	P07-retail-segmentation/
P08	P08-household-power/

P01 · Hourly bike-rental demand for fleet rebalancing

Question. Given hour, calendar and weather, how many bikes will be rented in the next hour — and which factors matter most?

Data. UCI Bike Sharing, `hour.csv` (17 379 rows, 1.1 MB). <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset> — browser download of the zip. CC BY 4.0.

Done. Cleaned features (cyclic encoding of hour/month, categorical weather), a leakage-free chronological split, linear regression and ridge/lasso baselines, one non-linear model (tree ensemble or MLP), RMSE and R^2 reported on the held-out period with a justified choice between them, and coefficient interpretation.

Further. Gradient descent implemented by hand and compared with the closed-form solution (Unit III); separate `casual` vs `registered` models; a peak-hour classifier with ROC/AUC.

Units. III, IV, V (+ VI optional). **COs.** CO2, CO3, CO4.

P02 · Next-day air quality for an Indian city

Question. Can yesterday’s pollutant readings and the calendar predict tomorrow’s PM2.5 (regression) and the AQI bucket (classification) well enough to issue a public warning?

Data. CPCB *Air Quality Data in India (2015–2020)*, use `city_day.csv` (~30 000 rows). <https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india> — Kaggle login + browser download. Pick one or two cities.

Done. Missing-data handling documented (this set has many gaps), lag/rolling features built without look-ahead, a **time-ordered** split, regression (linear → regularised) for PM2.5, a classifier for the AQI bucket with confusion matrix and ROC/AUC, and an honest comparison against the naïve “tomorrow = today” baseline.

Further. Multi-city model with city as a feature vs one model per city; season/festival effects; imbalance handling for the rare “Severe” class.

Units. IV, V, VI. **COs.** CO2, CO3, CO4.

P03 · Hotel-booking cancellations and the overbooking decision

Question. At booking time, which reservations will be cancelled — and at what probability threshold should the hotel overbook?

Data. Hotel Booking Demand (Antonio, de Almeida & Nunes), 119 390 rows. <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand> — Kaggle login + browser download.

Done. A leakage audit (columns such as `reservation_status` must be dropped — say why),

encoding of the mixed categorical features, stratified split, logistic regression, KNN, decision tree, random forest and one boosting model compared on ROC/AUC and precision–recall, then a threshold chosen for a stated cost of an empty room vs a walked guest.

Further. Resampling vs class weights for the imbalance; calibration curve; a per-hotel (city vs resort) comparison.

Units. IV, V (logistic), VI. **COs.** CO2, CO3, CO4.

P04 · Early warning of student dropout

Question. From data known at enrolment and after semester 1, which students are at risk of dropping out, and what would a fair early-warning rule look like?

Data. UCI *Predict Students’ Dropout and Academic Success*, 4 424 students $\times$ 36 features, three classes (dropout / enrolled / graduate), strongly imbalanced. <https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success> — browser download. CC BY 4.0.

Done. Multiclass classification with at least three model families (naïve Bayes, tree/ensemble, SVM or MLP), stratified cross-validation, macro-F1 and per-class recall reported and defended over plain accuracy, an “enrolment-only” model vs a “plus semester-1” model, and a feature-importance discussion that flags any demographic feature a university should *not* act on.

Further. Cost-sensitive thresholds; one-vs-rest ROC; a 2-class (dropout vs not) reformulation and whether it changes the ranking of models.

Units. IV, VI (+ V logistic). **COs.** CO1, CO2, CO3, CO4.

P05 · What makes a review negative? (text)

Question. Can a linear model on bag-of-words beat non-linear models on 50 000 movie reviews, and which words actually drive the prediction?

Data. IMDB 50K Movie Reviews, 50 000 labelled reviews, balanced. <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews> — Kaggle login + browser download (~66 MB).

Done. A text-cleaning pipeline (HTML stripping, tokenisation, stop-words — with an ablation showing whether each step helped), TF-IDF features, logistic regression and multinomial naïve Bayes and a linear SVM compared with accuracy and ROC/AUC, the top ± 20 weighted words shown and sanity-checked, and a learning curve (accuracy vs training size).

Further. Bigrams and `min_df` sweeps; error analysis of 30 misclassified reviews; hierarchical or k-means clustering of reviews using the TF-IDF vectors to find sub-genres.

Units. II, IV, V, VI (+ VII optional). **COs.** CO2, CO3, CO4.

P06 · Clothing-item recognition with classical ML

Question. How far can non-deep methods (PCA + KNN / SVM / random forest / a small MLP) get on 28×28 clothing images, and which classes are confused with which?

Data. Fashion-MNIST (Zalando), 70 000 images, 10 classes. <https://github.com/zalando-research/fashion-mnist> — browser download of the four gzip files, or the CSV version on Kaggle. MIT licence.

Done. A loader for the raw IDX files, pixel scaling, PCA with an explained-variance curve and a chosen number of components, at least four classifiers compared under one protocol on the official test split, per-class precision/recall and a confusion matrix with a written explanation of the shirt / T-shirt / pullover confusion, and a timing table (fit and predict seconds per model).

Further. Backpropagation MLP written with NumPy vs scikit-learn’s; k-means on PCA embeddings vs the true labels (adjusted Rand index); data augmentation by shifts.

Units. IV (dimensionality reduction), VI, VII optional. **COs.** CO1, CO2, CO3, CO4.

P07 · Customer segmentation for an online retailer

Question. Which distinct customer groups exist in two years of transactions, and can a classifier assign a *new* customer to a segment after their first few purchases?

Data. UCI Online Retail II, 1.07 M transactions, Dec 2009–Dec 2011 (43.5 MB xlsx). <https://archive.ics.uci.edu/dataset/502/online+retail+ii> — browser download. CC BY 4.0.

Done. Cleaning documented (cancellations, negative quantities, missing `CustomerID`), RFM features per customer (Recency, Frequency, Monetary), scaling justified, k-means with the elbow/silhouette choice of k , hierarchical agglomerative clustering with a dendrogram and a linkage comparison, DBSCAN with a discussion of what it calls noise, the segments named and described in business terms, then a classifier trained to predict the segment from early-purchase features with cross-validated accuracy.

Further. Year-1 clusters vs year-2 clusters — do customers migrate?; product-level clustering from the description text.

Units. IV, VI, VII. **COs.** CO1, CO2, CO3, CO4.

P08 · Household electricity: forecasting and daily-profile clustering

Question. Can the next hour’s household power draw be forecast from recent history, and what typical “kinds of day” does the household have?

Data. UCI Individual Household Electric Power Consumption, 2.07 M one-minute readings over 47 months (127 MB text, 20 MB zip). <https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption> — browser download. CC BY 4.0.

Done. Resampling from minutes to hours with the ~1.25 % missing rows handled and documented, lag/rolling and calendar features without look-ahead, a chronological split, linear/ridge regression vs a tree ensemble vs the persistence baseline on MAE and RMSE, then each day reshaped into a 24-value profile and clustered (k-means and HAC) with the cluster centroids plotted and interpreted (weekday/weekend, holiday, summer).

Further. Sub-metering (kitchen / laundry / heater) as targets; DBSCAN to find anomalous days; an MLP forecaster.

Units. IV, V, VII (+ VI ensembles). **COs.** CO2, CO3, CO4.

Coverage check

	P01	P02	P03	P04	P05	P06	P07	P08
Task family	regression	time-series reg. + clf	imbalanced binary	imbalanced multiclass	text	images	clustering → clf	time series + clustering
Unit IV preprocessing	✓	✓	✓	✓	✓	✓	✓	✓
Unit V regression	✓	✓	logistic	logistic	logistic			✓
Unit VI classification	opt.	✓	✓	✓	✓	✓	✓	opt.
Unit VII clustering					opt.	opt.	✓	✓
CO1				✓		✓	✓	
CO2	✓	✓	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓	✓	✓
Data size	1 MB	~3 MB	16 MB	0.5 MB	66 MB	30 MB	44 MB	20 MB zip

All eight need Unit IV plus at least one of V/VI/VII; none can be finished before Unit V is taught. None reuses a lab anchor dataset (housing, churn, stock, digits, Iris, spam, German credit, HAR, breast cancer, PlantVillage), so the project cannot be a re-submitted lab.

Effort band. P01 and P04 are the most contained (small, clean data); P07 and P08 carry the most data-wrangling. The “done” definitions are sized so that the wrangling-heavy themes ask for fewer model families.

Data routes

All eight links were checked on 2026-09-04 and resolve. UCI zips download directly; the Kaggle sets need a free account and the “Download” button; Fashion-MNIST is four gzip files linked from the GitHub README. Mirrors are placed on the LMS where licensing permits — check there first.